Tempe 1.22 Pavement Quality Index (PQI) Analysis Report

Oct 8, 2026 · @Simon

Summary

The bond will push Tempe's pavement score past the passing line quickly; but if nobody funds upkeep once the money runs out, the score will likely fall back below 70 around the 2036 deadline. The best-value approach is "mostly preventive upkeep, with some repaving."

Plain-language background:

- PQI is a "health score" for roads: 0 is the worst, 100 is brand new. The City's goal is a citywide average of at least 70 by June 2036.
- Today the citywide average is 65. Almost 1 of every 3 miles of road scores below 55 (31%).
- The City is using a bond approved by voters in 2024 to repair streets on a large scale over about 4 years.

Finding	Data behind it
1. Passing is not hard: the average will clear 70 by the end of 2027 at the latest	In all 7 test scenarios, PQI at end of 2027 is 74–79 (Model C; see the sensitivity table in Output)
2. Keeping it is hard: with no funding after the bond, the City will likely miss the deadline	In 6 of 7 scenarios, PQI in June 2036 is below 70 (52–69); base case 69.3
3. If only the average matters, "mostly upkeep" is cheapest	Holding 70 costs about $3.0M a year, 46% less than "fix the worst first" at $5.5M; savings are 37–51% in all 7 scenarios (Model D)
4. But upkeep alone lets bad roads grow, so the answer is a mix	With upkeep only, roads below 55 rise to about 24%; for the same $5.5M a year, a mix scores 2–5 points higher than "fix the worst first" with a similar share of bad roads (about 18%)

Dollar figures use 2016 unit prices from another city. They only compare strategies; they are not Tempe's actual budget.

Background: what Tempe is doing now

Tempe is borrowing a large sum to repair almost every street over about 4 years; but the money comes only once, and the City has not said what will pay for upkeep afterwards.

Where the PQI score comes from

- Every 3 years the City inspects each road segment for cracks, potholes and smoothness, then gives it a score from 0 to 100.
- The citywide average sat mostly between 56 and 60 for the past decade and rose to 65 at the end of 2025; the official status is yellow (Concerned).
- Official target (measure 1.22): a citywide average of at least 70 by June 30, 2036, met in neighborhoods of every demographic group.

What the bond program is

A bond works like a family borrowing from the bank to renovate the house now, then paying it back slowly over many years. The benefit is not waiting a decade to save up; the limit is that the money comes once, and it can usually be spent only on big projects such as repaving, not on yearly routine upkeep.

- Voters approved it on November 5, 2024. Road work is part of a $301 million bond question that also covers fire, police, sidewalks and storm drains; the road share was not published separately.
- The City says the work would take 11–15 years without the bond, and about 4 years (around 2028) with it.
- When asking for the bond, the City said nearly 600 miles of road were rated "fair to poor."

The City's plan

Two repair methods to know:

- **Mill and overlay**: scrape off the old top layer of asphalt and lay a new one. Like major surgery: it works well but is expensive.
- **Surface seal**: brush a protective coat onto a road that is still in decent shape, sealing small cracks. Like sunscreen: cheap, but it only helps roads that are still fairly good.

Item	Detail
Scope	About 90% of the City's 1,200 miles of road will be overlaid or sealed
Order	Start with the worst roads
Volume	FY2025–2028 plan of about 400 miles: 61% mill and overlay, 39% seals and other preservation treatments
Bond goal	Around 2028, every street above 82 ("excellent")
This year's supporting work (through June 2026)	Re-inspect 201 miles of local and collector streets north of US60; meet quarterly with water, power and gas utilities so new pavement is not dug up; update the database to an expected citywide average of about 64; build a system to track repair time and cost
The City's long-term view	Preserving good roads costs far less than fixing bad ones, so the same money covers 4 to 10 times more streets

The two targets are different: the official 70 is a citywide average with a 2036 deadline; the bond's 82 applies to every single street, by about 2028, and is a higher bar.

From background to problem

Observation	Data
Fixing the worst roads first during the bond makes sense, because there are so many	92.7 miles score below 40, or 19% of the network
Under the plan, the passing line will be crossed soon	Above 70 by the end of 2027 at the latest (all 7 scenarios)
After the money runs out, roads keep aging	Roads below 85 lose about 3 points a year (Model B)
The City says preservation is best value, but there is no long-term budget yet	The official page says 4–10 times; no post-2028 upkeep budget appears in public data

So the question is not "how do we pass?" but "after the bond, which upkeep approach keeps the gains for the least money?"

Problem Framing

The question is: after the bond, which upkeep approach costs the least while keeping the score above the passing line?

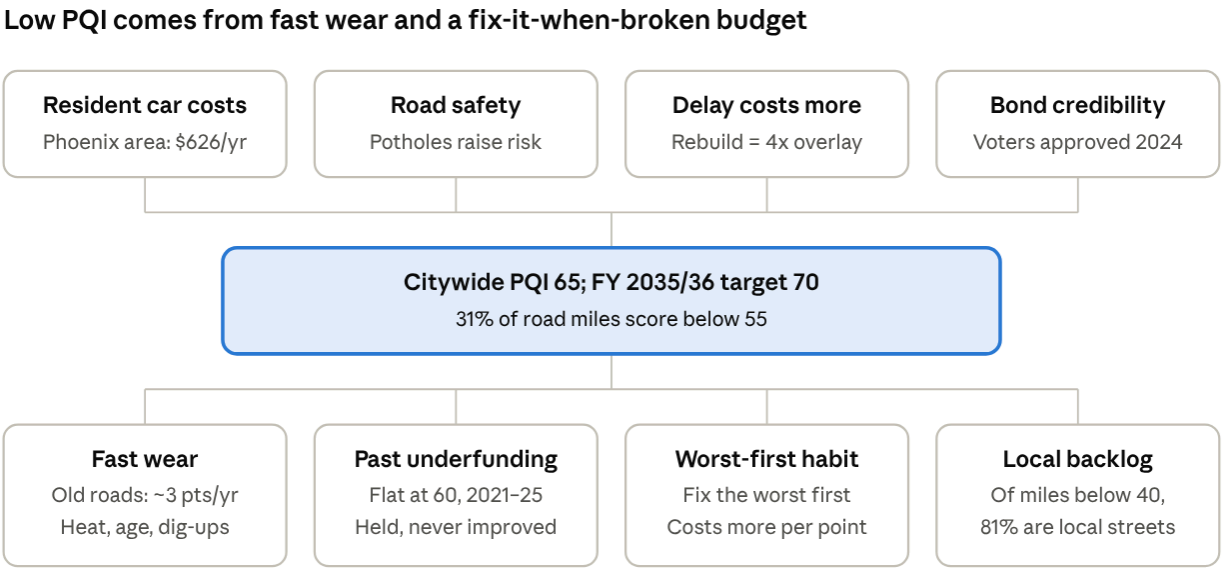
Problem Statement

> What post-bond pavement maintenance strategy would allow Tempe to keep its citywide PQI at or above 70 through the FY 2035/36 deadline and beyond, at the lowest annual cost?

In plain words: think of the City's roads as a set of teeth. The question is "how much should we spend each year on cleanings and how much on root canals, so the teeth stay healthy for the least money?"

The sentence deliberately does not assume the answer: "mostly preventive upkeep" is a result of the analysis, not a premise written into the question.

Problem tree



Problem tree · 4 effects, 1 core problem, 4 root causes

The top row shows how poor roads affect residents and the City; the bottom row shows four causes. Each cause can be checked with data, and they do not overlap (MECE):

Cause	How it is checked	Result
Fast wear	Model B (City segment database)	Roads below 85 lose 3.0 points a year (mean of 15 groups)

Cause	How it is checked	Result
Past underfunding	1.22 citywide history	56–60 every year from 2010 to 2025; trend slope only +0.14 points a year
Worst-first habit	Model D (cost per point gained)	Sealing costs about $8,600 per mile per point; repaving a 60-point road about $22,600
Local street backlog	June 2026 segment data	Of the 92.7 miles below 40, 74.9 miles (81%) are local streets

Six questions

Question	Answer
What is this really about?	How the City should split its road money so roads stay good to drive on for the least cost
Why change?	PQI stayed at 56–60 for a decade (official status Concerned, yellow); bond money runs out around 2028 and no budget is set for afterwards
What is the central issue?	Once the bond lifts the score, which upkeep approach holds 70 for the least money each year
Who are the players?	See the stakeholder table below
What are the boundaries?	Only the four City-managed road types (arterial, collector, local, local-industrial; 483.9 miles); excludes private roads and alleys (1.8 miles) and state highways
What does success look like?	Citywide PQI at or above 70 every year from FY 2035/36 onward

SMART check

Item	Plain-language check	Result
Specific	Asks about "after the bond" and "which upkeep approach"; subject and timing are clear	Pass
Measurable	Judged by two numbers: citywide average PQI and dollars per year	Pass

Item	Plain-language check	Result
Actionable	The answer is "seal X miles and repave Y miles a year," which the City can schedule directly	Pass
Relevant	Maps directly to official measure 1.22 and the 2024 bond	Pass
Time-bound	Meet the target by June 2036, checked through 2040	Pass

Stakeholders

Who has the most power to decide and who cares the most sets how we communicate with them (Power/Interest Grid).

Stakeholder	Role in this issue	What they care about most	How to engage
City Council	Approves the budget	Whether the bond delivers, and where money comes from afterwards	Manage Closely
Public Works pavement management	Does the repairs	Which streets to fix first when money is limited	Manage Closely
Transportation department	Owns traffic safety	Arterial condition and crashes	Keep Satisfied
Residents and drivers	Use the roads; also voters	Vehicle wear, closures during work	Keep Informed
Businesses	Use the roads	Customers kept away during construction	Keep Informed
Water, power and gas utilities	Dig up pavement to lay lines	Construction schedules	Keep Satisfied

Data

Most of the data comes straight from the City's open data portal; outside sources only fill in costs and background.

In plain words, we have four ledgers: one records "the citywide average each year," one records "each road's score today," one records "each road's score when last inspected and what is left by 2025," and one records "which roads the City plans to fix."

Dataset	In plain words	Expert detail
1.22 PQI Average Citywide (summary)	The citywide average each year	14 yearly points (2010–2026); the table lists a target of 70 by 2030, which conflicts with the official FY 2035/36 date; this report follows the official document
Pavement Quality Index Segments	Each road's score today, length and type	Snapshot of 2026-06-03; 485.7 centerline miles; fields RM_MI (miles), rm_func (road class), per_pqi_yr1 (PQI)
1.22 PQI Road Segment (detail)	Each road's score when last inspected, and its projected 2025 score	5,312 segments; inspected 2023 (1,971), 2024 (2,339), 2025 (939), earlier 41, no year 22; fields MEASURED_PQI, CURRENT_PQI, YEAR, CLASS
Planned Road Resurfacings	Which roads the City plans to fix, when, and how	FY2025–2028; 13 treatment types; total length 2,111,554 ft (about 400 miles)

Outside sources:

Source	Used for
1.22 official performance dashboard	Target (≥ 70 by June 2036), status, this year's strategies, "4 to 10 times"
Tempe Pavement Management Program	Inspection every 3 years; three treatment types
Arizona PBS (June 2025)	1,200 miles; every street above 82 by about 2028; worst first
City of Littleton maintenance costs	2016 bid prices per lane-mile: surface seal $35,000, mill and overlay $320,000, reconstruction $1.3M
TRIP report (KJZZ coverage)	Phoenix-area drivers pay an extra $626 a year because of rough roads

How the data was pulled: through ArcGIS server-side aggregation (sums and averages by road class, year and score band), not row-by-row download; the models therefore use grouped numbers. Found but not yet used: Traffic Count Segments (24-hour traffic volumes since 2016).

Analysis and models

Four models, each feeding the next; each is explained first in plain words, then with the numbers an expert needs.

Model	Plain-language picture	Method	Question it answers
A. Trend baseline	Draw a line through past scores and see where it heads	Linear regression	Without the bond, what is the score in 2036?
B. Wear rate	Measure the shoe sole: how much wears off each year	Linear regression through the origin	How many points does a road lose a year?
C. Network simulation	Each year, take points off every road and add points to repaired ones; compute the citywide average	Binned condition simulation	Under each approach, what is the score from 2026 to 2040?
D. Cost optimization	Try every mix of "how much to repave, how much to seal" and keep the cheapest	Grid search	What is the least it costs to stay above the passing line?

A. Trend baseline: what happens if nothing changes

In plain words: scores over the past 15 years are almost a flat line; at that pace, without the bond, the City would reach only about 61 by 2036.

- Data: citywide averages for 2010–2025, 13 yearly points (the 2026 value of 65 already includes bond work, so it is left out).
- Result: slope +0.14 points a year, $R^2 = 0.22$, 2036 forecast 61.4.
- Reading: R^2 is low because the line is flat. This model only shows that "without the bond, 70 is out of reach"; it is not used to forecast the bond's effect.

B. Wear rate: how many points a road loses each year

In plain words: the City records two numbers for each road segment—"its score when inspected" and "its projected score in 2025." Subtract and divide by the years in between, and you get points lost per year. Result: roads below 85 lose about 3 points a year.

$$\text{annual loss}_{c,y} = \frac{\overline{PQI}^{\,inspected}_{c,y} - \overline{PQI}^{\,2025\ projected}_{c,y}}{2025 - y}$$

c is the road class and y the inspection year. Splitting by class × inspection year gives 16 groups:

Road class	Inspection year	Mean at inspection	Mean projected for 2025	Points lost per year
Arterial	2023	70.7	64.7	3.00
Arterial	2020	66.9	51.0	3.18
Collector	2024	68.3	66.1	2.15
Collector	2023	65.8	58.9	3.42
Collector	2021	38.1	24.2	3.47
Collector	2020	96.1	91.1	1.00
Collector	2018	66.6	38.8	3.97
Local	2024	62.9	60.4	2.49
Local	2023	55.6	49.1	3.25
Local	2021	41.0	27.6	3.36
Local	2019	63.9	51.2	2.12
Local	2018	56.5	30.5	3.71
Local	2014	64.4	26.6	3.44
Local-industrial	2024	71.7	69.8	1.91
Local-industrial	2023	63.6	57.6	3.01
Local-industrial	2021	69.6	60.6	2.26

- The 15 groups starting below 85: mean loss 2.98 points a year, median 3.18, range 1.91–3.97.
- Cumulative loss regressed on elapsed years through the origin: slope 3.32 points a year, $R^2 = 0.93$.

- New pavement (85 and above) has only 1 group (Collector 2020, 96.1 → 91.1), losing about 1 point a year. The sample is too small, so the sensitivity test also runs 2 points a year.
- Key limit: the "2025 projected" value comes from the City's system applying its own wear curve; it is not a fresh inspection. Model B therefore recovers the City system's assumption. Once the City's 2026 re-inspection of 201 miles is published, old and new inspections can be compared directly.

C. Network simulation: the score each year under each approach

In plain words: put the City's 485.7 miles of road into score bins from 0 to 100. Every year, each bin loses points (roads age); if there is money, some roads gain points (they get fixed). Repeat until 2040 and compute the citywide average each year.

Rule	Setting	Basis
Aging	Roads below 85 lose 3 points a year; 85 and above lose 1	Model B
Mill and overlay	Starts from the lowest-scoring roads; repaired roads become 95	Local, collector and industrial roads re-inspected in 2025 averaged 95.3–96.1
Surface seal	Applied to roads at 60–85; adds 10 points (cap 92)	Assumption: a seal adds about 3–5 years of life at about 3 points a year
Bond-period volume	Second half of 2026: 64.9 miles; 2027: 101.2; 2028: 73.9; 61% overlay, 39% seal	Planned Road Resurfacings (129.8 miles for all of 2026; the snapshot is from June, so half is assumed to remain)
Starting point	Even spread within each score band	June 2026 segment snapshot

Validation:

- The model starts at 66.2, 1–2 points above the official current value of 65 and the City's expected 64 for June 2026; the sensitivity test therefore includes a "start 2 points lower" case.
- Running the City's own plan, the model reaches 82.8 at the start of 2029, in line with the City's aim of "every street above 82 around 2028." But an average of 83 is not every street above 82: the model still shows 7.8% of miles below 55 at that point.

D. Cost optimization: the least it costs to stay above the line

In plain words: feed every combination of "miles repaved and miles sealed per year" into Model C, keep the ones that pass in every year from 2029 to 2040, then pick the cheapest.

- Search range: overlay 0–80 miles (steps of 0.5) and seal 0–200 miles (steps of 2).
- Constraint: citywide average ≥ 70 in every year 2029–2040; a second version also caps the share of miles below 55.

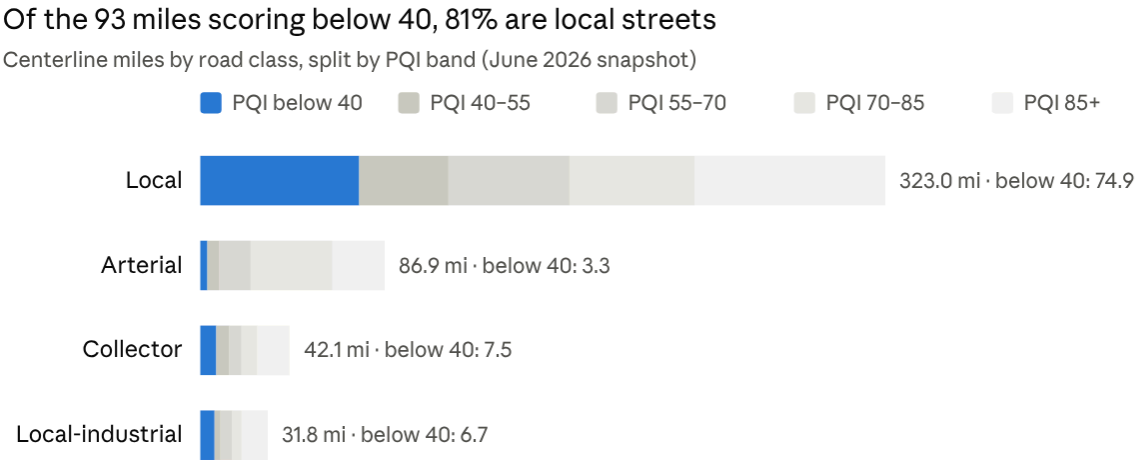
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\text{annual cost} = (\text{overlay miles} \times 320{,}000 + \text{seal miles} \times 35{,}000) \times 2.47
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2.47 is the lane factor (the City's 1,200 miles ÷ 485.7 centerline miles in the data). Per centerline mile, that is about $86,500 to seal and $790,600 to repave.

Output

The analysis produces three things: where roads are worst today, how the score moves over the next 15 years, and how much to trust those forecasts.

1. Today: bad roads are concentrated on local streets

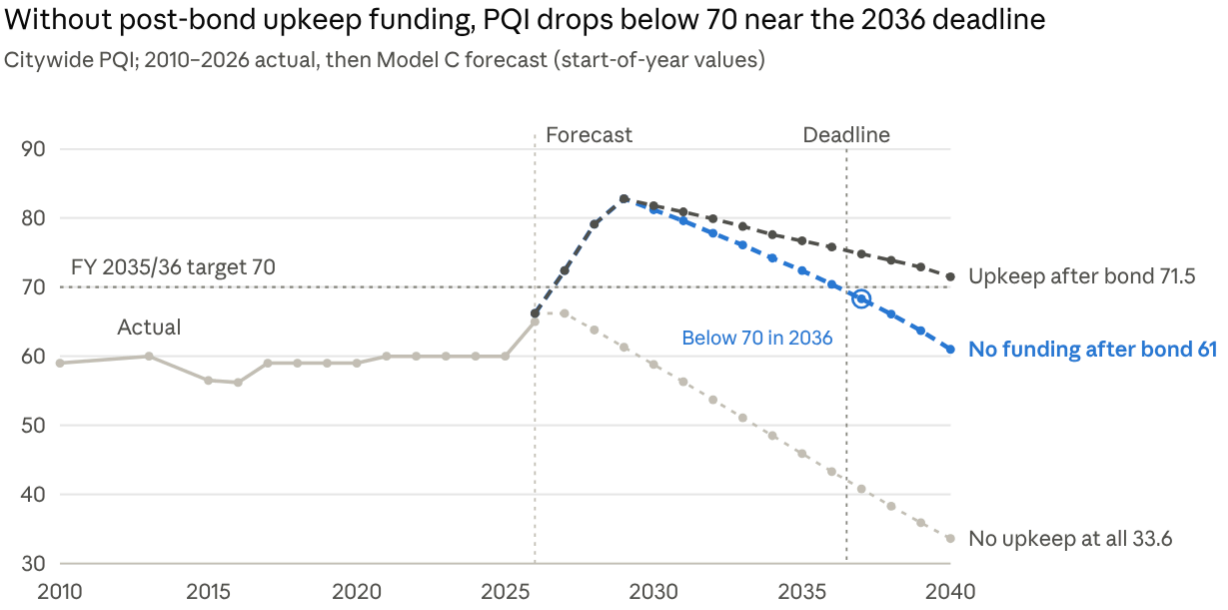


Pavement Quality Index Segments · June 2026 snapshot, excluding private roads and alleys (1.8 miles)

In plain words: blue is road scoring below 40. The local-street bar is the longest and has the most blue.

- All 485.7 miles: below 40, 19.1%; 40–55, 11.9%; 55–70, 17.2%; 70–85, 22.5%; 85 and above, 29.3%.
- Share below 55: arterial 10.2%, collector 32.2%, local 36.2%, local-industrial 29.9%.
- Mean segment score by class: arterial 74.2, collector 71.1, local 64.1, local-industrial 67.8.

2. The future: three paths after the bond



1.22 PQI Average Citywide (2010–2026) + Model C simulation (2026–2040, base case)

In plain words: the solid gray line is the past and dashed lines are forecasts. The blue line is "stop spending when the bond runs out," which drops below 70 around the 2036 deadline; the dark gray line is "spend about $3M a year on upkeep afterwards," which holds through 2040.

- With no further spending, the yearly drop in the citywide average grows: 1.6 points in 2029, 2.7 points in 2039. Freshly repaired roads age slowly at first (1 point a year), then faster once they fall below 85 (3 points a year).
- "Fix the worst first" and "mostly upkeep" nearly overlap on the average (76.5 vs 75.8 at the start of 2036), so only one is drawn; they differ in cost and share of bad roads (see Business insights).

3. How reliable are these forecasts: 7 scenarios

In plain words: the forecasts rest on a few assumptions. We pushed each one in the bad direction to see whether the findings flip.

Scenario	End of 2026	End of 2027	Peak, start of 2029	No further spending: June 2036	No further spending: year it drops below 70	Hold 70: mostly upkeep (per year)	Hold 70: fix worst first (per year)	Savings
Base case	72.4	79.1	82.8	69.3	2036	$3.0M	$5.5M	46%
Start 2 points lower (≈ City's expected 64)	70.6	77.4	81.2	67.1	2035	$3.9M	$6.7M	42%

Scenario	End of 2026	End of 2027	Peak, start of 2029	No further spending: June 2036	No further spending: year it drops below 70	Hold 70: mostly upkeep (per year)	Hold 70: fix worst first (per year)	Savings
Only 1/4 of 2026 work left after June	69.4	76.4	80.5	66.0	2034	$4.1M	$7.1M	42%
New roads lose 2 points a year	72.4	78.7	81.7	61.9	2033	$6.7M	$13.8M	51%
Old roads lose 2.5 points a year	72.4	79.3	83.3	72.7	2038	$1.6M	$2.8M	44%
Old roads lose 3.5 points a year	72.4	78.9	82.3	65.7	2034	$4.7M	$7.5M	37%
Most conservative (all bad assumptions at once)	67.5	74.1	77.3	52.1	2031	$10.2M	$18.2M	44%

Dollar figures use 2016 unit prices from another city. Three findings hold in every scenario:

1. **Passing by the end of 2027 is certain**: all 7 scenarios score 74 or higher. Passing by the end of 2026 is not certain (67.5–72.4).
2. **Without further spending, the score is not held**: 6 of 7 scenarios are below 70 in June 2036; only the optimistic "old roads lose 2.5 points" case holds (72.7).
3. **Mostly upkeep is always cheaper**: savings of 37–51% in all 7 scenarios.

Business insights and recommendations

Do not pick a single method: spending most of the money on keeping fair roads in shape, and a smaller share on rebuilding the worst ones, gets the best result for the same budget.

Insight 1: passing is easy; keeping it is hard

In plain words: the bond is like a big spring cleaning—the house is clean fast; but if nobody keeps cleaning, it is dirty again in a few years.

- At the end of 2027, all 7 scenarios score 74 or higher.
- With no spending after the bond, 6 of 7 scenarios fall below 70 by June 2036; the base case is 69.3.

Insight 2: if only the average counts, "mostly upkeep" is cheapest

In plain words: a cleaning costs far less than a root canal. Spending a little to protect a road while it is still decent beats waiting for it to fail and then doing major repairs.

Method	Cost per mile	Points gained	Cost per mile per point
Surface seal (roads at 60–85)	$86,500	+10 (assumed)	$8,600
Repave a road at 30	$790,600	+65 (30 → 95)	$12,200
Repave a road at 60	$790,600	+35 (60 → 95)	$22,600

- To hold 70: "mostly upkeep" costs $3.0M a year (0.5 miles repaved + 30 miles sealed); "fix the worst first" costs $5.5M a year (7 miles repaved): 46% less, and 37–51% less in all 7 scenarios.
- The City itself says preservation covers 4–10 times more streets, so the direction matches; our figure is more conservative because it still keeps some repaving.

Insight 3: but upkeep alone lets bad roads grow

In plain words: cleanings without fillings keep the average looking fine, but the decayed teeth keep getting worse.

Approach (base case)	Cost per year	Average, start of 2036	Below 55, start of 2036	Average, start of 2040	Below 55, start of 2040
No further spending	$0	70.4	27.6%	61.0	32.7%
Mostly upkeep (0.5 mi overlay + 30 mi seal)	$3.0M	75.8	24.2%	71.5	23.8%
Fix the worst first (7 mi overlay)	$5.5M	76.5	17.6%	71.8	16.9%
Mix (1.5 mi overlay + 50 mi seal)	$5.5M	78.3	18.9%	76.4	17.7%

For the same $5.5M a year, the mix scores 1.8–4.6 points higher on average than "fix the worst first," with only about 1 percentage point more bad road. So the best answer is not one or the other, but a mix.

Insight 4: local streets drive the score; arterials drive what residents feel

In plain words: the citywide average is weighted by road length. Local streets are the most miles, so they move the score the most; but most cars drive on arterials.

- Local streets are 66.5% of all miles and 81% of the miles below 40.

- Arterials average 74.2, with only 10.2% below 55.
- TRIP estimates Phoenix-area drivers pay an extra $626 a year in repairs and wear because of rough roads.
- So when ordering projects, traffic volume should count alongside the score.

Insight 5: protecting new roads is cheaper than fixing old ones

In plain words: digging up a freshly paved road to lay pipes is like cutting a hole in a brand-new shirt.

- One of the City's official strategies this year is to meet quarterly with utility companies so newly paved roads are not dug up.
- The sensitivity test shows that if new roads age at 2 points a year instead of 1 (for example, after being dug up), the cost of holding 70 rises from $3.0M to $6.7M a year, 123% more.

Recommendations

Recommendation	Data behind it
1. Set a post-bond upkeep budget before 2028: about $3.0M a year to hold the average only, or about $5.5M to also control bad roads (2016 prices)	Insights 1, 2, 3
2. Use a mix: most money on sealing roads at 60–85, a smaller share on repaving the worst	For the same $5.5M, average 1.8–4.6 points higher (Insight 3)
3. Track two KPIs together: citywide average PQI and the share of miles below 55 (31% today)	Watching only the average misses bad roads rising from 17.6% to 24.2% (Insight 3)
4. Add traffic volume when ordering projects	Local streets drive the score; arterials carry the traffic (Insight 4)
5. Make utility coordination a standing rule to protect bond-funded new pavement	New roads aging twice as fast raises cost by 123% (Insight 5)

Limitations

The three core findings hold in all 7 scenarios; but dollar amounts and years are estimates and should be updated once the City's own data is available.

Limitation	In plain words	How much it matters	How to close it
Grouped data only	We see "how many miles fall in each score band," not each road	The model assumes an even spread within each band	Rebuild Model C with segment-level data
Wear rate is a system projection	"About 3 points a year" is the City system's assumption, not a fresh inspection	Tested at 2.5–3.5 points for old roads; findings hold	Compare once the 2026 re-inspection of 201 miles is published
Only 1 group for new roads	"New roads lose 1 point a year" rests on one example	At 2 points, cost rises 123%, but mostly-upkeep still saves 51%	Same as above
Seal adds 10 points (assumed)	Not a Tempe measurement	Affects the cost of the upkeep option	Request before-and-after inspections of sealed roads
Unit prices are from another city in 2016	Littleton, Colorado bid prices; amounts run low	Used only to compare strategies; ratios are more reliable than amounts	The City is building its own cost baseline
Lane factor 2.47 is an estimate	Assumes the City's 1,200 miles are lane-miles	Affects amounts, not ratios	Confirm with the City
Planned lengths may double-count	Divided roads may be counted in both directions, overstating bond-period scores	Tested in the "only 1/4 of work left" case	Check the plan layer
New roads being dug up	Utility work makes new roads age faster	Tested in the "new roads lose 2 points" case	Track the City's coordination results
Traffic volume not yet joined	We do not yet know which bad roads carry the most cars	Affects project order, not the citywide average	Join Traffic Count Segments

Sources

- Tempe City Council Strategic Priorities and Performance Measures (November 2025)
- 1.22 Pavement Quality Index official dashboard
- Council Priorities and Strategic Plan
- 1.22 PQI Average Citywide (summary)
- Pavement Quality Index Segments
- 1.22 PQI Road Segment (detail)
- 1.22 PQI Road Segment data dictionary
- Planned Road Resurfacings
- Traffic Count Segments
- Tempe Pavement Management Program
- Massive efforts being made for Tempe streets (Arizona PBS, June 2025)
- Refresh Tempe
- Tempe asking voters for $581.5 million to fix roadways (Mary Scott Nabers, September 2024)
- Street Maintenance Definitions and Costs (City of Littleton)
- Phoenix-Area Road Conditions Cost Drivers $2,000 Annually (KJZZ / Fronteras)